

A Pathological Diagnosis Method Combining Image-Text Large Model and Rule-Based Reasoning

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Abstract—Biopsy is considered as the “gold-standard” for disease diagnosis. Cooperative reasoning from multiple perspectives of the induction of the morphology and structure of cells and tissues, and the deduction of the cross disciplinary knowledge of clinical and pathological is needed. Multimodal image-text-pair based large models have improved the ability of generalization for computational pathological analysis, but lack of capability of reasoning for diagnosis. To address the above limitation, a pathology-base diagnosis method combining large models with rule-based reasoning is proposed in this paper to consider performance of generalization and reasoning comprehensively. The reasoning rules, summarized by experts to reflect medical domain knowledge, serve as the core components for diagnostic decisions. Specifically, clinical practice guidelines are transformed to rules of *Datalog* which is a reasoning engine based on first-order predicate logic. Features extracted from pathological analysis using pre-trained large image-text models are treated as facts of *Datalog* for rule-fact matching during reasoning. In order to verify the usability of our proposed method, a case study of a downstream task of liver fibrosis staging is demonstrated, and comparison is conducted among different studies.

Keywords—multimodal pre-trained large model, convolutional neural network, computational pathology, medical image processing, computer-aided diagnosis

I. INTRODUCTION

Digital pathology refers to computer-stored image of tissues and cells around the extracellular matrix that is enlarged under a microscope and formed after staining with H&E etc. The storage size of whole pathology slide image (WSI) with a magnification of 40×(40 times) may exceed 1GB. It can reflect the morphology, texture, arrangement, and other features of diseased tissues or cells more clearly. For many diseases, the results of pathology

slide analysis are considered as the “gold-standard” for diagnostic and therapeutic decisions.

Traditionally, pathological analysis relies on pathologists to observe the cell morphology and tissue structure under the microscope with human efforts, and to give analysis reports based on personal experience and expertise. There exist the following difficulties:

(1) The high dependence of disease diagnosis and treatment on pathological analysis results in a heavy workload for pathologists. With the public's demand for high-quality medical services, the workload of pathological diagnosis has increased, and the volume of clinical pathology slide samples in various hospitals has grown exponentially. The increased pathological samples and the quick feedback speed have caused many pathology departments to operate under high load.

(2) The high accuracy requirements of the results of pathological slide analysis raise higher requirements for the personal experience and expertise of pathologists. Pathologists assist clinicians in making judgments and decisions based on observing the morphological changes of diseases in cells and tissues. They need to master the knowledge of basic medicine, clinical medicine, anatomy, histology, biochemistry, molecular biology, immunology and other multi-disciplinary knowledge.

In view of the above challenges in pathological analysis, computational pathology combines digital pathology with artificial intelligence technology.

One of the cores of computational pathology is the use of deep learning methods for medical image analysis. At present, many researchers have designed AI models using labelled data which are collected from specific types of diseases, and achieved significant image feature recognition performance.

However, generalization is an important obstacle to the application of these models.

The emergence of large language models provides a way to solve model generalization. On the basis of pre-trained large language models, massive medical image-text pairs used to fine-tune existing multimodal large models for medical image retrieval and classification. However, when used in pathological diagnosis analysis, the ability of image-text pairs to express domain knowledge is weak, resulting in poor reasoning ability.

Based on the above analysis, this paper designs a pathological assisted diagnosis method combining large models with rule engines to enhance the reasoning ability and interpretability of image-text large models. The detailed content of this paper includes:

(1) A pathological diagnosis method combining Multimodal Large Model and Rule-based Reasoning is designed. The zero-shot capability of multimodal large models is used for feature classification of pathological slices; Disease diagnosis knowledge associated with downstream tasks are converted into rules of the rule-based reasoning engine; The results of the image-text large model serve as facts of the rule-based reasoning engine. Two-level reasoning is achieved based on deep learning and rule reasoning.

(2) In addition to the way of tiling, Whole Slide Images are also divided into standard-sized tiles through the way of overlap and rotation to avoid the global characteristics of pathological sections will not be destroyed in data preprocessing stage.

(3) Pathology-based disease diagnosis will involve quantitative values of morphological features and domain rules. These diagnostic rules combine different pathological concepts, values, and their quantitative comparison relationships. This paper adopts a reasoning machine based on first-order predicate logic to implement diagnostic rule reasoning after image-text pairing.

The rest of this paper is organized as follows. Section II analyzes the related work from the viewpoints of multimodal large model applications and pathology-based diagnosis decision supporting. Section III presents the general framework and details of our approach to enhance the reasoning ability of image-text large models in downstream tasks. Section IV illustrates a case study of staging diagnosis of liver fibrosis. Finally, Section V summarizes the paper and briefly describes future research directions.

II. RELATED WORK

As a gold standard, the goal of CP (Computational Pathology) is to assist doctors in clinic decision making. Difficulties exist in both image feature extraction and diagnosis inference.

With the application of AI in image processing, different deep learning models have been used to automatically extract pathological image features to reduce missed or false detections of lesions caused by insufficient experience or fatigue of physicians. Although WSIs (Whole Slide Images) images preserve rich pathological features for pathologists to perform hierarchical feature recognition based on pyramid structured data models, the huge amount of data also brings computational

pressure to deep learning. When building the break cancer dataset for grand challenge in [1], WSIs are divided into patches for annotation and model training. Currently, there are open-source tools, such as *openslide*, that can be download from <https://openslide.org> for flat patch segmentation in C or Python. It is ease of implement flat cutting of WSIs. However, it is possible to divide a lesion into different patches, leading to lesion feature missing. Therefore, in this research, in addition to flat way, overlay and rotation splitting are also adopted.

When detecting features of WSI patches, the main methods are based on CNNs. 2D or 3D convolutional networks already have high accuracy in medical image recognition, such as pulmonary nodules detection in [2]. In order to train CNN models, substantial amounts of labeled data are needed. For pathological images, the annotation workload exceeds imagination. So that weak supervised deep learning methods combined with data augmentation, such as in [3], are often used in CP for disease grading or lesion detection.

However, deep learning models trained using specific dataset are usually weak in generalization. In order to perform multiple tasks, image-text large models are explored through pre-training and transfer learning. Both research in [4] and [5] build pathological oriented datasets to fine tune existing image-text large model through contrastive learning to form pathological oriented pre-training large model. Researchers in [4] believe that social networks contain valuable pathological information. So that they crawl *Twitter* and collect pathological images and their description to form graph-text pairs. Researchers in [5] download medical journal papers, extract pathological images and their captions to form graph-text pairs.

Due to the high dependence on prompts for LLM (Large Language Model), how to generate correct prompts for downstream tasks still require exploration. Moreover, for decision-making, the image-text pair match model lacks reasoning ability.

Multimodal data fusion is a trend in subtype classification or survival time prediction based on medical image analysis. Studies, such as in [6], use CNN networks to gain image feature vectors, MLP networks to gain clinic data feature vectors, fully connected layer to classify the jointed multimodal features. The advantage of this kind of multimodal data fusion method is that it can leverage the advantages of medical big data, while the disadvantage is that it cannot be explained and existing medical knowledge, such as diagnosis and treatment guidelines, is not utilized.

Data-knowledge dual driven approach is another kind of method for multimodal data fusion, which has received research attentions recently. Abductive reasoning method proposed in [7] uses induction results of data-driven inference phase as input of rule-based inference machine for verification. Due to the richer knowledge representation capabilities of knowledge graphs, many researchers have designed inference methods that combine neural networks with knowledge graphs[8]. In these methods, neural networks are used to enhance the construction ability of knowledge graphs, as well as knowledge graphs are used to guide parameter optimization of neural networks. Because of the complexity of graph computation, the combination of neural networks with knowledge graphs is still need exploration.

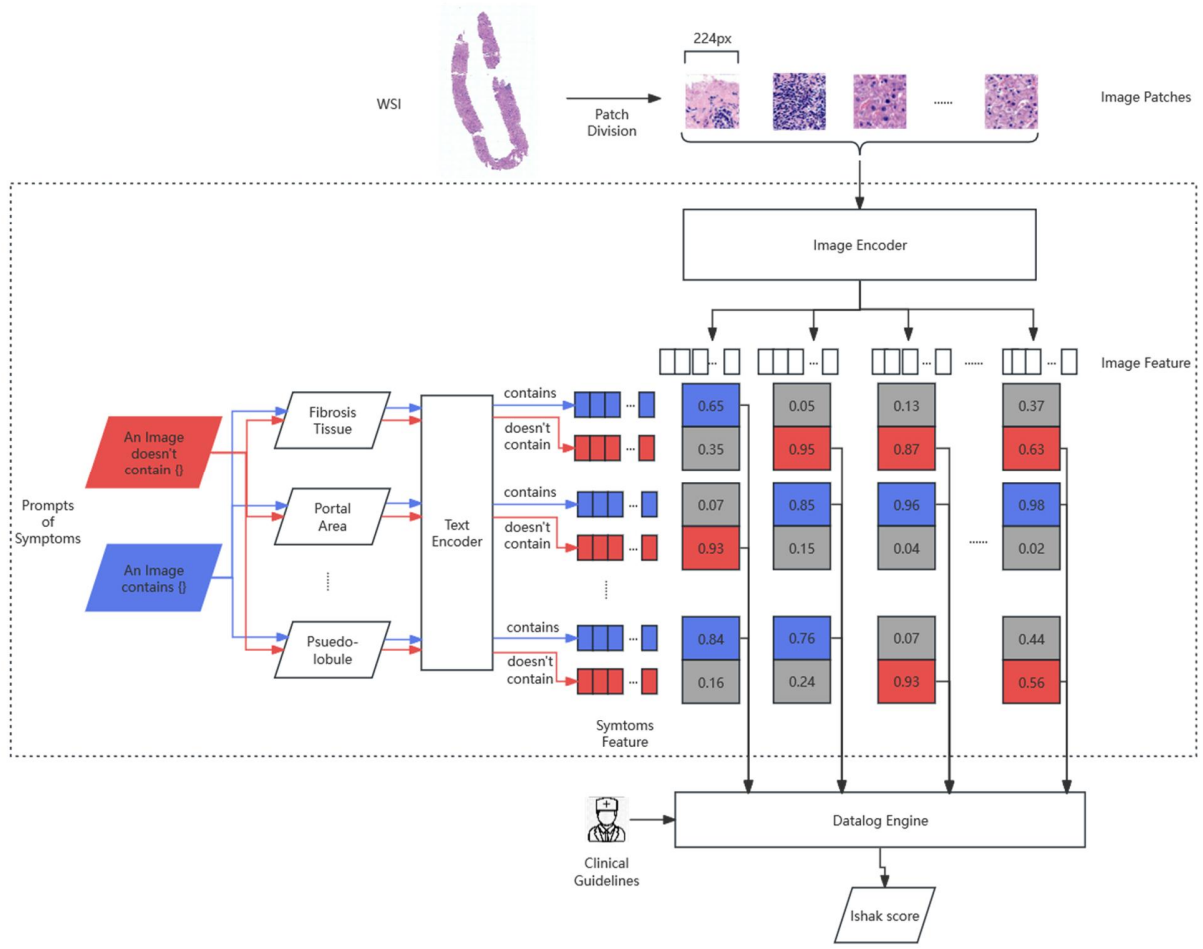


Fig. 1. Framework of pathological diagnosis method of combining image-text large model and rule-based reasoning

The above analysis shows that in the field of pathological diagnosis, there are limitations in specific-task oriented CNN models. In diagnostic and treatment guidelines, quantitative physiological indicators are often encountered. Therefore, in this paper method that combines first-order predicate logic with LLM is proposed to enhance the effect of diagnosis decision support.

III. PATHOLOGICAL DIAGNOSIS METHOD OF COMBINING IMAGE-TEXT LARGE MODEL AND RULE-BASED REASONING

A. Framework of the pathological diagnosis method

The main idea of our method is to use a rule-based reasoning engine to achieve explainable pathological diagnosis inference, as shown in Fig. 1. It mainly includes 3 components: WSI image splitting, patch image-text matching, and diagnosis rule reasoning. Pathological images are analyzed using deep learning model. Diagnosis is inferred using rule-based reasoning engine reflecting medical knowledge.

The dotted box in Fig. 1 shows the image-text pre-trained large model, which automatically extracts pathological image features through deep learning. The main purpose of using the

"image-text pairs" large model to extract patch image features is to deal with the zero-shot problem that is prone to occur in pathological analysis.

The results of the "image-text pairs" large model are treated as the facts of the rule-based reasoning engine, while treatment guidelines or expert experience are pre-defined as the rules of the reasoning engine.

Considering that quantitative variables are involved in diagnosis deduction, first-order predicate logic, which includes *rules* and *fact*, is used for rule-based reasoning.

Facts compose of *atoms*. *Atoms* are defined in format of $P(x_1, x_2, \dots, x_n)$, where P represents a logical relationship among X_i , and X_i is a variable or constant represents medical concept in pathological diagnosis.

Rules are defined in format of $H \leftarrow B_1, B_2, \dots, B_n$, where H is the head *atom* and B_i is the body *atom*. If all body *atoms* are *true*, then the head *atom* is assigned as *true*, standing that a new *fact* is deduced.

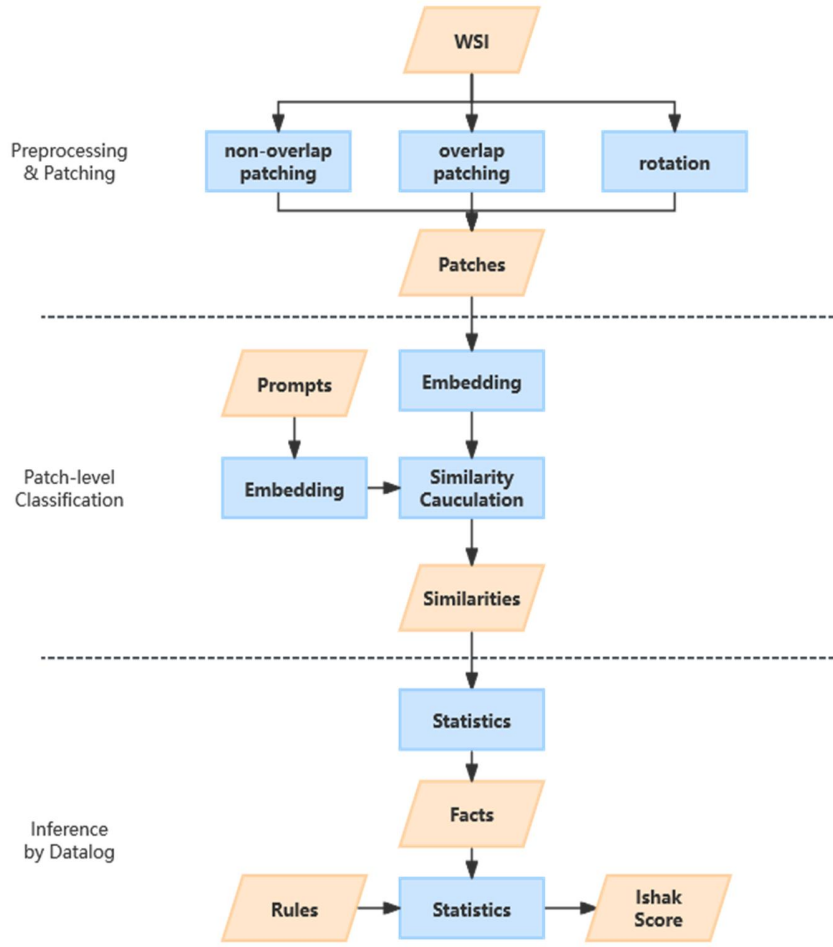


Fig. 2. Flowchart of pathological diagnosis prototype combining image-text pre-trained large model with rule-based reasoning

Before WSIs are inputted into pre-trained large models, in order to recognize their histopathology features as comprehensively as possible through image-text pair matching, the WSI images should be split to unified patches. In Fig. 1, three different splitting types are designed: (1)tile cutting that can maintain the position relations among different patches; (2)rotated cutting that can adapt to different angles of lesion morphology, and (3)overlap cutting that prevent small size of lesions from being divided into different patches.

B. Design of the pathological diagnosis prototype

In order to verify the proposed pathological diagnosis method, a prototype is designed in this paper. The flowchart of the prototype is shown in Fig. 2. It mainly consists of three steps: image pre-processing, patch-level image-text matching, and diagnostic rule inference.

(1)In image pre-processing step, the foreground regions of the WSIs are extracted and cut into small patches. For the foreground extraction, we uses the *Otsu* algorithm included from *Python (version 3.8.18)* library of *opencv-python (version 4.8.1.78)* to binarize the WSIs.

TABLE I. PSEUDO CODE 1: OVERLAP PATCH CUTTING

Input: WSI, WSI foreground mask, patch size, proportion of overlap area, minimum proportion of ROI area in the patch
Output: overlap divided patches

```

for x in range(0, WSI.width) do:
  for y in range(0, WSI.height) do:
    patch = WSI[x : x + patch_size][y : y + patch_size]
    for each point in patch do:
      if point belong to ROI_region:
        ROI_pixels++
      if ROI_pixels / patch.area > ROI_proportion:
        result.append(patch)
    x += patch_size * (1 - overlap_proportion)
    y += patch_size * (1 - overlap_proportion)
return result

```

Sliding windows are used to cut the WSIs into patches in ways of tiling, overlapping with certain proportion, and rotation with certain angles. The pseudo code of overlapping is shown in Table 1. Patches with certain proportion of *ROI* region are saved.

The width and height of the rotated image is calculated with the following equation.

$$\begin{aligned} \text{rotated_width} &= \text{WSI.height} * \sin(\text{angle}) + \text{WSI.width} * \cos(\text{angle}) \\ \text{rotated_height} &= \text{WSI.height} * \cos(\text{angle}) + \text{WSI.width} * \sin(\text{angle}) \end{aligned}$$

(2) In patch-level image-text matching step, the pre-trained image-text large model *PLIP* in [4], which could be downloaded from <https://tinyurl.com/webplip.PLIP>, is imported to calculate the cosine similarity between patch features and prompts. Because *PLIP* is fine-tuned on multimodal large model *CLIP*, in order to execute *PLIP* script code, *CLIP* needs to be imported (https://colab.research.google.com/github/openai/clip/blob/main/notebooks/Interacting_with_CLIP.ipynb).

(3) In diagnosis inference step, *pyDatalog* (version 0.17.4) which is a Datalog (Database logic) inference engine tool based on first order predicate logic is deployed to perform rule-fact matching.

Before inputting the results of the image-text matching into the inference engine, pathological features of the patches need to be summed up while considering the cutting method and regions of the patches in WSI-level. The pseudo code of WSI-level feature statistic is shown in Table 2.

TABLE II. PSEUDO CODE 2: WSI-LEVEL FEATURE STATISTIC

```

Input: S symptoms' classification result by PLIP of patches
Output: S facts of pyDatalog
for each symptom do:
    symptom_heat_map = numpy.zeros(WSI.shape)
    for each patch p do:
        if {symptom}.classification_result of patch is positive:
            symptom_heat_map[area corresponding to the patch] = 1
        num_labels, labels = cv2.connectedComponents(symptom_heat_map)
        {symptom}.number = num_labels - 1
        {symptom}.area = symptom_heat_map.countPixels(pixel == 1)
    if portal_area_with_fibrosis.area / WSI.area > 0.5:
        Fibrosis = 'most portal area'
    if portal_area_with_fibrosis.area == 0:
        Fibrosis = 'no'
    return Fibrosis, Fibrosis_septa.number, P_P.number, P_C.number,
    Nodules.number

```

IV. CASE STUDY AND DISCUSSION

A. Case study: pathological diagnosis for liver fibrosis staging

Liver fibrosis is a kind of abnormal proliferation of liver connective tissue which can be caused by many diseases such as alcoholic liver, fatty liver, viral infections, et.al. If left untreated, liver fibrosis can develop into cirrhosis, which can be life-threatening. So correct pathological diagnosis for liver fibrosis staging helps doctors to deliver timely treatment.

Taking diagnosis task for liver fibrosis staging as example, the proposed method is demonstrated including WSIs pre-processing, patch-level image-text matching and diagnosis inference.

(1) WSIs pre-processing: As shown in Fig.3 WSIs are cutting into patches in ways of tiling (Fig.3(a)), overlapping (Fig.3(b)) and rotating (Fig.3(c)) after foreground segmentation is performed. Patches that contain ROIs are retained.

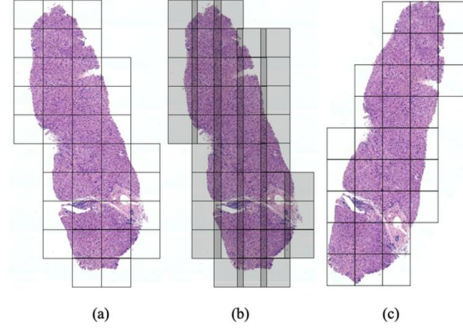


Fig. 3 Example of cutting (a) tile (b) overlap (c) rotation

The liver patches are inputted into pre-trained large image-text matching model *PLIP* for local pathological feature recognition. The prompts for *PLIP* are mapped from the atomic variables of rules in *pyDatalog* inference machine.

(2) In this case study, as showed in Fig. 4, *Ishak* score guideline which is one of accepted liver fibrosis staging standards, is transformed to rules of *pyDatalog*.

```

from pyDatalog import pyDatalog

pyDatalog.create_terms('Ishak_score, Result, Fibrosis, Fibrosis_septa, P_P, P_C, Nodules')
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 0) <- (Fibrosis == 'no')
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 1) <- (Fibrosis == 'some portal area') & (Fibrosis_septa == 0)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 2) <- (Fibrosis == 'most portal area') & (Fibrosis_septa <= 1)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 3) <- (Fibrosis == 'most portal area') & (Fibrosis_septa <= 2)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 4) <- (Fibrosis == 'some portal area') & (Fibrosis_septa >= 4) & (P_P == 1) & (P_C == 1)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 5) <- (Fibrosis == 'most portal area') & (Fibrosis_septa >= 4) & (P_P == 1) & (P_C == 1)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 5) <- (P_P == 1) & (Nodules >= 1) & (Nodules <= 3)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 5) <- (P_C == 1) & (Nodules >= 1) & (Nodules <= 3)
(Ishak_score(Fibrosis, Fibrosis_septa, P_P, P_C, Nodules) == 6) <- (Nodules > 3)

print(Ishak_score('most portal area', 2, 0, 0, 0) == Result)

```

Fig. 4 Rules in *pyDatalog* transformed from *Ishak* score guideline

According to the definition of rules in *pyDatalog*, we extract the atomic variables, which are *portal area*, *fibrous septs*, *fibrosis tissue*, *portal to portal bridging*, *portal to central bridging*, and *pseudo-lobule* to form a set of prompts of *PLIP*. For each patch, each prompt will be checked, as shown in Fig.5.

```

an Image doesn't contains portal area : 0.35853463
an Image contains portal area : 0.64146537
group result: an Image contains portal area

-----
an Image doesn't contains fibrous septa : 0.7862814
an Image contains fibrous septa : 0.21371865
group result: an Image doesn't contains fibrous septa

-----
an Image doesn't contains fibrosis tissue : 0.86472774
an Image contains fibrosis tissue : 0.13527222
group result: an Image doesn't contains fibrosis tissue

-----
an Image contains portal to portal bridging : 0.2557464
an Image contains portal to central bridging : 0.835431214
an Image doesn't contains portal to portal bridging, nor portal to central bridging : 0.78882237
group result: an Image doesn't contains portal to portal bridging, nor portal to central bridging

-----
an Image doesn't contains pseudo-lobule : 0.728361
an Image contains pseudo-lobule : 0.27969897
group result: an Image doesn't contains pseudo-lobule

```

Fig. 5. *PLIP* image-text matching output

The outputs of *PLIP* with thresholds exceeding certain values will be treated as the inputs of the *pyDatalog* after statistic, which is shown in Fig.6.

(3) According to the statistic result of the *PLIP* outputs of the patches divided from a WSI, and rules pre-defined according to *Ishak* score guideline, the *pyDatalog* gives the liver fibrosis staging score of the WSI, as shown in Fig.7.

```

image0 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image doesn't contains pseudo-lobule']
image1 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image2 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image3 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image4 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image5 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image6 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image7 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image8 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']
image9 ['an image contains portal area', 'an image doesn't contains fibrous septa', 'an image doesn't contains fibrosis tissue', 'an image doesn't contains portal to portal bridging, nor portal to central bridging', 'an image contains pseudo-lobule']

```

Fig. 6 Statistic results of local features of patches

```

fibrosis:      most portal area
fibrous septa: 2
p.p:          1
p.c:          1
pseudo-lobule: 1
Result
-----
5

```

Fig. 7 Inference result of *pyDatalog* of a WSI

B. Comparison and discussion

The simulation process of liver fibrosis staging show that when multidisciplinary knowledge are needed in fields such as pathological diagnosis, hybrid reasoning methods combing neural network reasoning, such as image-text large model, with rule-based inference, such as *Datalog* model, would better to be used to realize perceptual learning and explainable reasoning to solve complex and difficult decision-making problems.

TABLE III. COMPARISON WITH OTHER SIMILAR METHODS

		Reference [5]	Reference [6]	Reference [8]	Our Method
Function	Goal	Foundation Model	CT Diagnosis	Medical Question and Answer	Pathology Diagnosis
Structure	Data Preprocess	WSI to Patch	No	Voice to Text	WSI to Patch
	Feature Extraction	ViT& GPT-style Transformer	CNN&MLP	NLP	<i>PLIP</i>
	Inference	Visual Text Match	Joint ViT and MLP	Knowledge Graph	<i>Datalog</i> Inference
Behaviour	Simplicity	✓	✓	✗	✓
	Generalization	✓	✗	✓	✗
	Interpretability	✗	✗	✓	✓

Compared with researches in [5] [6] and [8] in the relative field of AI-aided disease diagnosis. As both in [5] and our method, in data preprocess, WSIs are usually divided into patches for image feature extraction. Due to the explicit knowledge representation, the interpretabilities in [8] and our method are relative strong. But knowledge graph construction in [8] is more complex. The generalizations of foundation model in [5] and question/answer system in [8] are high based on pre-trained large model. Method in [6] is specifically designed for interstitial lung disease. In our method, the WSIs analysis is general because that pre-trained large model is involved, but the rules of *Datalog* are transformed from special diseases. So the generalization of our method is medium compared with others.

V. CONCLUSION

Computational pathological analysis is widely used in lesion classification and disease staging for such as tumors and fibrosis. In order to reduce the workload and improve the efficiency of

pathological analysis, AI methods are introduced to detect lesions automatically. However, due to the heterogeneity of different diseases, deep-learning methods encounter the difficulties in task generalization and cross discipline reasoning for diagnosis. Various AI methods need to collaborate with each other to complete inter-disciplinary knowledge collaboration.

In this paper we propose a data and knowledge dual driven method of pathological analysis combining image-text large model *PLIP* together with first predicate logic based reasoning *pyDatalog* to use pathology image perception and medical knowledge analysis comprehensively. In order to make good use of the zero-shot image feature recognition ability of *PLIP* for pathological patches, we design various ways to cut WSIs into patches with unified size. In order to enhance the interpretability of diagnosis reasoning, statistic algorithm is designed to summarize features recognized by *PLIP* to facts of *pyDatalog*. Rules of *pyDatalog* reasoning engine are pre-defined according to clinical diagnosis guidelines.

There are still limitations to our proposed method. The accuracy of the nidus recognition of patches still need to be improved. In additionally, there is a gap between the image-text matching results and facts used in the reasoning engine. The future work will focus on the patch size definition and prompt selection to improve accuracy of pre-trained large model, and the fusion of image perception and medical knowledge.

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